

POC Proposal: Dynamic Word Template Filler

Executive Summary

A web-based application that enables users to upload Word document templates using simple placeholder syntax and dynamically generate filled documents through an intuitive form interface.

Business Need

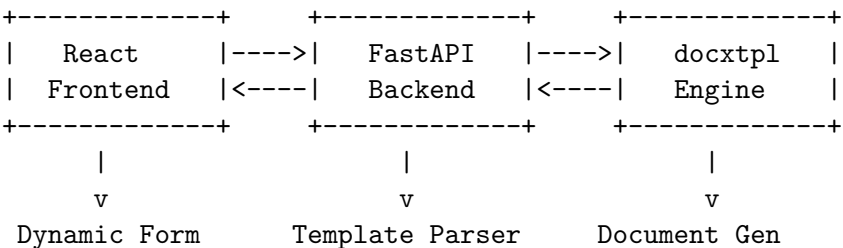
Organizations frequently need to generate customized Word documents from templates (contracts, invoices, reports). Current solutions require technical knowledge of complex templating systems. This POC demonstrates a user-friendly approach using intuitive placeholder syntax.

Proposed Solution

Template Syntax Support

Syntax	Purpose	Example
{field}	Simple text placeholder	{name}, {email}
{#section}...{/section}	Conditional content	{#disclaimer}Legal text{/disclaimer}
{#list}...{/list}	Repeatable sections	{#products}{name} - {price}{/products}

System Architecture



User Interface Wireframe

Key Features

1. **Drag & Drop Upload** - Simple template upload with validation
2. **Auto-Detection** - Automatically extracts fields, conditionals, and lists
3. **Dynamic Forms** - Generates appropriate input controls per field type
4. **Repeatable Sections** - Add/remove items for list-type content
5. **Conditional Sections** - Toggle optional content via checkboxes
6. **Instant Generation** - Download filled document immediately

· DocxTpl Template Filler

Upload Template 

Drop .docx file here or click to browse

· Template Fields Detected

Simple Fields

Name:

Email:

Company:

Optional Sections

☒ Include Disclaimer

☐ Include Terms and Conditions

Products (Repeatable)

Item 1	
Product Name:	Widget A
Price:	\$10.00
Remove	
Item 2	
Product Name:	Widget B
Price:	\$25.00
Remove	
+ Add Product	

Generate Document

Figure 1: UI Wireframe

Technical Stack

Component	Technology
Frontend	React + TypeScript + Vite
Backend	Python + FastAPI
Template Engine	python-docx-template (docxtpl)
Document Format	OOXML (.docx)

Success Criteria

- ☐ Parse templates with `{field}`, `{#conditional}`, `{#list}` syntax
 - ☐ Generate dynamic form UI from parsed template
 - ☐ Support add/remove for repeatable sections
 - ☐ Generate filled documents preserving original formatting
 - ☐ Handle empty/optional fields gracefully
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Timeline Estimate

Phase	Duration
Backend Parser & Generator	2 days
Frontend Dynamic Form	2 days
Integration & Testing	1 day
Total	5 days

Document generated from implemented POC solution